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ELECTRO-OPTICAL DEVICE AND ELECTRONIC APPARATUS

Abstract

An electro-optical device includes a substrate, a transistor including a gate electrode and a semiconductor layer, a scanning line, a first light shielding portion disposed in a first direction with respect to the semiconductor layer, and a second light shielding portion disposed in a second direction with respect to the semiconductor layer, the first light shielding portion includes a first portion that is extends from a first insulating layer between the scanning line and the gate electrode to a second insulating layer between the semiconductor layer and the second light shielding portion, an intermediate layer is disposed between the first portion and the second light shielding portion to be in contact with the first portion and the second light shielding portion, and the intermediate layer includes a layer having an insulating property and contains a material different from materials of the first insulating layer and the second insulating layer.

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Background/Summary

[0001] The present application is based on, and claims priority from JP Application Serial Number 2024-035132, filed Mar. 7, 2024, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to an electro-optical device and an electronic apparatus.

2. Related Art

[0003] For electronic apparatuses such as projectors, for example, electro-optical devices such as liquid crystal display devices whose optical characteristics can be changed for each pixel are used. As an example of the electro-optical device, a display device disclosed in JP-A-2015-7806 is known.

[0004] The display device disclosed in described in JP-A-2015-7806 includes a thin film transistor (TFT) including a semiconductor film and a gate electrode disposed in a layer above the semiconductor film, and a scanning line disposed in a layer above the gate electrode. The gate electrode and the scanning line are electrically coupled via a contact hole. Further, in the display device, the scanning line provided in a layer above the semiconductor film is used as a light shielding layer.

[0005] In devices of the related art, a scanning line provided in a layer above a semiconductor film is used as a light shielding layer, but there is a problem that a light shielding property with respect to the semiconductor film is not sufficient only with this.

SUMMARY

[0006] According to an aspect of the present disclosure, an electro-optical device includes a substrate, a transistor including a gate electrode disposed in a first direction with respect to the substrate and a semiconductor layer disposed between the substrate and the gate electrode, a scanning line disposed in the first direction with respect to the gate electrode and electrically coupled to the gate electrode, a first light shielding portion disposed in the first direction with respect to the semiconductor layer and overlapping the semiconductor layer in plan view when viewed in the first direction, and a second light shielding portion disposed in a second direction opposite to the first direction with respect to the semiconductor layer and overlapping the semiconductor layer in plan view, in which a potential of the first light shielding portion and a potential of the second light shielding portion are different from each other, the first light shielding portion includes a first portion that is provided on both sides of the semiconductor layer in a width direction and extends from a first insulating layer between the scanning line and the gate electrode to a second insulating layer between the semiconductor layer and the second light shielding portion, an intermediate layer is disposed between the first portion and the second light shielding portion to be in contact with the first portion and the second light shielding portion, and the intermediate layer includes a layer having an insulating property and contains a material different from materials of the first insulating layer and the second insulating layer.

[0007] According to an aspect of the present disclosure, an electronic apparatus includes an electro-optical device, and a control unit configured to control an operation of the electro-optical device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is a plan view of an electro-optical device according to an embodiment.

[0009] FIG. **2** is a cross-sectional view taken along line A-A of the electro-optical device shown in FIG. **1**.

[0010] FIG. **3** is an equivalent circuit diagram showing an electrical configuration of an element substrate of FIG. **1**.

[0011] FIG. **4** is a plan view showing a portion of the element substrate in a display region in FIG. **2**.

[0012] FIG. **5** is a cross-sectional view taken along line A**1**-A**1** in FIG. **4**.

[0013] FIG. **6** is a cross-sectional view taken along line A**2**-A**2** in FIG. **4**.

[0014] FIG. **7** is a plan view showing a transistor in FIG. **5**.

[0015] FIG. **8** is a plan view of a second light shielding portion shown in FIG. **5**.

[0016] FIG. **9** is a plan view of a first light shielding portion and a second light shielding portion shown in FIG. **5**.

[0017] FIG. **10** is a plan view of the first light shielding portion and the second light shielding portion shown in FIG. **5**.

[0018] FIG. **11** is a cross-sectional view taken along line A**3**-A**3** in FIG. **10**.

[0019] FIG. **12** is a cross-sectional view taken along line A**4**-A**4** in FIG. **10**.

[0020] FIG. **13** is a plan view of an intermediate layer shown in FIG. **12**.

[0021] FIG. **14** is a diagram showing a method of manufacturing the intermediate layer shown in FIG. **12**.

[0022] FIG. **15** is a diagram showing a method of manufacturing the intermediate layer shown in FIG. **12**.

[0023] FIG. **16** is a cross-sectional view showing a method of manufacturing a first concave portion.

[0024] FIG. **17** is a plan view showing a method of manufacturing the first concave portion.

[0025] FIG. **18** is a cross-sectional view showing a method of manufacturing a second concave portion.

[0026] FIG. **19** is a plan view showing a method of manufacturing the second concave portion.

[0027] FIG. **20** is a plan view showing a method of manufacturing a third concave portion.

[0028] FIG. **21** is a diagram showing a method of manufacturing a first light shielding portion.

[0029] FIG. **22** is a cross-sectional view showing an intermediate layer in a second embodiment.

[0030] FIG. **23** is a cross-sectional view showing the intermediate layer in the second embodiment.

[0031] FIG. **24** is a plan view showing a second layer of the intermediate layer shown in FIG. **22**.

[0032] FIG. **25** is a perspective view showing a personal computer which is an example of an electronic apparatus.

[0033] FIG. **26** is a plan view showing a smartphone which is an example of an electronic apparatus.

[0034] FIG. **27** is a schematic diagram showing a projector which is an example of an electronic apparatus.

DESCRIPTION OF EMBODIMENTS

[0035] Hereinafter, preferred embodiments of the present disclosure will be described with reference to the accompanying drawings. In the drawings, dimensions or scales of respective portions may be appropriately different from actual ones, and there are portions schematically shown to facilitate understanding. Further, the scope of the present disclosure is not limited to these forms unless there is a particular statement that limits the present disclosure in the following description.

1. Electro-Optical Device

1A. Basic Configuration

[0036] FIG. **1** is a plan view of an electro-optical device **100** according to an embodiment. FIG. **2** is a cross-sectional view taken along line A-A of the electro-optical device **100** shown in FIG. **1**. An opposing substrate **3** is not shown in FIG. **1**. Further, hereinafter, for convenience of description, an X-axis, a Y-axis, and a Z-axis orthogonal to each other will be used as appropriate. Further, one direction along the X-axis is referred to as an X**1** direction, and a direction opposite to the X**1** direction is referred to as an X**2** direction. Similarly, one direction along the Y-axis is referred to as a Y**1** direction, and a direction opposite to the Y**1** direction is referred to as a Y**2** direction. One direction along the Z-axis is referred to as a Z**1** direction, and a direction opposite to the Z**1** direction is referred to as a Z**2** direction. The Z**1** direction is equivalent to a “first direction”, and the Z**2** direction is equivalent to a “second direction”.
[0037] Further, in this specification, “an element β on an element α” means that the element β is located on the side above the element α. Thus, “an element (on an element α” includes not only a case where the element β is in direct contact with element α, but also a case where the element α and the element β are separated from each other. Further, “electrical coupling” between an element α and an element β includes not only a configuration where the element α and the element β are electrically coupled by being directly bonded to each other, but also a configuration where the element α and the element β are electrically coupled indirectly through another conductive material.
[0038] The electro-optical device **100** shown in FIGS. **1** and **2** is a transmissive electro-optical device using an active matrix drive scheme. The electro-optical device **100** includes an element substrate **2**, an opposing substrate **3**, a frame-shaped sealing member **4**, and a liquid crystal layer **5**. As shown in FIG. **2**, the element substrate **2**, the liquid crystal layer **5** and the opposing substrate **3** are arranged in this order in the Z**1** direction. Viewing from the Z**1** direction or Z**2** direction, which is a direction in which these overlap, is referred to as “plan view”. Further, although the shape of the electro-optical device **100** shown in FIG. **1** is a rectangular shape in plan view, the shape may have a polygonal shape or a circular shape other than a rectangular shape.
[0039] The element substrate **2** shown in FIG. **2** includes a first substrate **21** having light transmittance, a stacked body **22** having light transmittance, a plurality of pixel electrodes **25** having light transmittance, and a first orientation film **29** having light transmittance. The first substrate **21** is equivalent to a “substrate”. The first substrate **21**, the stacked body **22**, the plurality of pixel electrodes **25**, and the first orientation film **29** are stacked in this order in the Z**1** direction. In addition, “light transmittance” means transmittance of visible light and preferably indicates that the transmittance to visible light is 50% or more. In addition, as will be described later in detail, the element substrate **2** includes a first light shielding portion **6** having a light shielding property shown in FIGS. **5** and **6**, a second light shielding portion **210**, and a third light shielding portion **7** having a light shielding property. “Light shielding property” means a light shielding property against visible light, and preferably means that the transmittance of the visible light is less than 50%, and more preferably 10% or less.
[0040] The first substrate **21** is a flat plate having light transmittance and an insulating property, and is constituted by a glass substrate or a quartz substrate, for example. The stacked body **22** includes a plurality of insulating films having light transmittance. In addition, the stacked body **22** is provided with various wiring lines and the like. The pixel electrode **25** is used for applying an electric field to the liquid crystal layer **5**. The pixel electrode **25** includes a transparent conductive material such as indium tin oxide (ITO), indium zinc oxide (IZO), and fluorine-doped tin oxide (FTO). Although not shown in the drawings, the element substrate **2** includes a plurality of dummy pixel electrodes surrounding the plurality of pixel electrodes **25** in plan view. In addition, the first orientation film **29** has light transmittance and an insulating property. The first orientation film **29** aligns liquid crystal molecules in the liquid crystal layer **5**. The first orientation film **29** is disposed to cover the plurality of pixel electrodes **25**. The material of the first orientation film **29** is polyimide, silicon oxide, and the like.

[0041] The opposing substrate **3** is disposed to face the element substrate **2**. The opposing substrate **3** includes a second substrate **31** having light transmittance, an inorganic insulating layer **32** having light transmittance, a common electrode **33** having light transmittance, and a second orientation film **34** having light transmittance. In addition, although not shown in the drawings, the opposing substrate **3** includes a light shielding parting that surrounds the plurality of pixel electrodes **25** in plan view.
[0042] The second substrate **31**, the inorganic insulating layer **32**, the common electrode **33**, and the second orientation film **34** are stacked in this order in the Z**2** direction. The second substrate **31** is a flat plate having light transmittance and an insulating property, and is constituted by a glass substrate or a quartz substrate, for example. The inorganic insulating layer **32** has light transmittance and an insulation property, and is made of an inorganic material containing silicon, such as silicon oxide. The common electrode **33** is an opposing electrode disposed to face the plurality of pixel electrodes **25** through the liquid crystal layer **5**. The common electrode **33** is used for applying an electric field to the liquid crystal layer **5**. The common electrode **33** has light transmittance and conductivity. The common electrode **33** contains, for example, a transparent conductive material such as ITO, IZO, and FTO. The second orientation film **34** has light transmittance and an insulating property. The second orientation film **34** aligns liquid crystal molecules in the liquid crystal layer **5**. The material of the second orientation film **34** is polyimide, silicon oxide and the like, for example.
[0043] The sealing member **4** is disposed between the element substrate **2** and the opposing substrate **3**. The sealing member **4** is formed using, for example, an adhesive containing various curable resins such as an epoxy resin. The sealing member **4** may include a gap material made of an inorganic material such as glass.
[0044] The liquid crystal layer **5** is disposed in a region surrounded by the element substrate **2**, the opposing substrate **3**, and the sealing member **4**. The liquid crystal layer **5** is an electro-optical layer whose optical characteristics change depending on an electric field. The liquid crystal layer **5** includes liquid crystal molecules having positive or negative dielectric anisotropy. An orientation of the liquid crystal molecules changes depending on a voltage applied to the liquid crystal layer **5**.
[0045] As shown in FIG. **1**, a plurality of scanning line drive circuits **11**, a signal line drive circuit **12** and a plurality of external terminals **13** are disposed at the element substrate **2**. Some of the plurality of external terminals **13** are coupled to a wiring (not shown) routed from the scanning line drive circuit **11** or the signal line drive circuit **12**. Further, the plurality of external terminals **13** include a terminal to which a constant potential Vcom is applied. The terminal is electrically coupled to the common electrode **33** of the opposing substrate **3** through a wiring line and a conductive material (not shown). In this manner, the constant potential Vcom is supplied to the common electrode **33**.
[0046] The electro-optical device **100** includes a display region A**10** in which an image is displayed, and a peripheral region A**20** located outside the display region A**10** in plan view. A plurality of pixels P disposed in a matrix are provided in the display region A**10**. The plurality of pixel electrodes **25** are disposed in a one-to-one relationship with the plurality of pixels P. The common electrode **33** described above is provided in common for the plurality of pixels P. Further, the peripheral region A**20** surrounds the display region A**10** in plan view. The scanning line drive circuit **11** and the signal line drive circuit **12** are disposed in the peripheral region A**20**.
[0047] In the present embodiment, the electro-optical device **100** is of a transmissive type. More specifically, as shown in FIG. **2**, after light LL is incident on the opposing substrate **3**, the light LL is modulated before being emitted from the element substrate **2**, whereby an image is displayed. Light having been incident on the element substrate **2** may be modulated before being emitted from the opposing substrate **3**, whereby an image is displayed.
[0048] Further, the electro-optical device **100** is applied to, for example, a display device that performs color display, of a personal computer, a smartphone, or the like, which will be described

below. When the electro-optical device **100** is applied to the display device, a color filter is appropriately used for the electro-optical device **100**. Further, the electro-optical device **100** is applied to, for example, a projection type projector, which will be described later. In this case, the electro-optical device **100** functions as a light valve. In this case, the color filter is omitted from the electro-optical device **100**.

1B. Electrical Configuration of Element Substrate **2**

[0049] FIG. **3** is an equivalent circuit diagram showing an electrical configuration of the element substrate **2** of FIG. **1**. As shown in FIG. **3**, the element substrate **2** includes a plurality of transistors **23**, n scanning lines **241**, m signal lines **242** and n constant potential lines **243**. n and m are integers equal to or greater than **2**. The transistor **23** is arranged in a manner corresponding to each intersection of the n scanning lines **241** and the m signal lines **242**. Each of the transistors **23** is a Thin Film Transistor (TFT) that functions as a switching element, for example. Each of the transistors **23** includes a gate, a source, and a drain.

[0050] Each of the n scanning lines **241** extends in the X**1** direction, and the n scanning lines **241** are disposed at equal intervals in the Y**1** direction. Each of the n scanning lines **241** is electrically coupled to the gates of a plurality of corresponding transistors **23**. The n scanning lines **241** are electrically coupled to the scanning line drive circuit **11** shown in FIG. **1**. Scanning signals G**1**, G**2**, . . . , and Gn are supplied line-sequentially from the scanning line drive circuit **11** to the 1 to n scanning lines **241**.

[0051] Each of the m signal lines **242** shown in FIG. **3** extends in the Y**1** direction, and the m signal lines **242** are arranged at equal intervals in the X**1** direction. The m signal lines **242** are respectively electrically coupled to the sources of the plurality of corresponding transistors **23**. The m signal lines **242** are electrically coupled to the signal line drive circuit **12** shown in FIG. **1**. Image signals S**1**, S**2**, . . . , and Sm are supplied in parallel from the signal line drive circuit **12** to the 1 to m signal lines **242**.

[0052] The n scanning lines **241** and the m signal lines **242** shown in FIG. **3** are electrically insulated from each other and disposed in a grid pattern in plan view. A region surrounded by two adjacent scanning lines **241** and two adjacent signal lines **242** corresponds to the pixel P. The transistor **23**, the pixel electrode **25** and a capacitive element **24** are provided for each pixel P. The pixel electrode **25** is provided in a one-to-one relationship for the transistor **23**. Each pixel electrode **25** is electrically coupled to the drain of the corresponding transistor **23**.

[0053] The n constant potential lines **243** are extended in the X**1** direction, and the n constant potential lines **243** are arranged at equal intervals in the Y**2** direction. Further, the n constant potential lines **243** are electrically insulated from the n scanning lines **241** and the m signal lines **242**, and are disposed at intervals from these lines. A constant potential Vcom is applied to each constant potential line **243**. Each of the n constant potential lines **243** is electrically coupled to one of the two electrodes of the corresponding capacitive element **24**. Each capacitive element **24** is a storage capacitor for holding the potential of the pixel electrode **25**. The capacitive element **24** is provided in a one-to-one relationship for the transistor **23**. In addition, the other of the two electrodes of each capacitive element **24** is electrically coupled to the corresponding pixel electrode **25**. Therefore, the constant potential Vcom is applied to one electrode of the capacitive element **24**, and the other electrode is electrically coupled to the drain of the transistor **23**.

[0054] When the scanning signals G**1**, G**2**, . . . , and Gn sequentially become active, and the n scanning lines **241** are sequentially selected, the transistors **23** coupled to the selected scanning line **241** are turned on. Then, the image signals S**1**, S**2** . . . and Sm with values corresponding to the gradation to be displayed through the m signal lines **242** are taken by the pixel P corresponding to the selected scanning line **241**, and applied to the pixel electrode **25**. Thereby, a voltage corresponding to the gradation to be displayed is applied to a liquid crystal capacitor formed between the pixel electrode **25** and the common electrode **33** in FIG. **2**, and the orientation of the liquid crystal molecules changes in accordance with the applied voltage. In addition, the applied voltage is held by the capacitive element **24**. Light is modulated due to such changes in the orientation of liquid crystal molecules, making gradation display possible.

1C. Structure of Portion of Element Substrate **2**

[0055] FIG. **4** shows a portion of the element substrate **2** in the display region A**10** of FIG. **2**. As shown in FIG. **4**, the display region A**10** includes a plurality of opening regions A**11**, and a light shielding region A**12**. The plurality of opening regions A**11** are disposed in a matrix in plan view. The shape of the light shielding region A**12** in plan view is a frame shape located between the plurality of opening regions A**11**. Each opening region A**11** is a region in which the pixel electrode **25** is disposed, and is a region through which light passes. On the other hand, the transistor **23** is disposed in the light shielding region A**12**. Although not shown in FIG. **4**, various wiring lines such as the scanning lines **241**, the signal lines **242**, and the constant potential lines **243** shown in FIG. **3**, and the capacitive element **24** are disposed in the light shielding region A**12**.

[0056] FIG. **5** is a cross-sectional view taken along line A**1**-A**1** in FIG. **4**. FIG. **6** is a cross-sectional view taken along line A**2**-A**2** in FIG. **4**. As shown in FIGS. **5** and **6**, the element substrate **2** includes the first substrate **21**, which is a "substrate", and the stacked body **22**. The stacked body **22** includes a plurality of insulating layers **221**, **222**, **223**, **224**, **225**, **226**, **227**, **228** and **229**. The insulating layers **221**, **222**, **223**, **224**, **225**, **226**, **227**, **228** and **229** are stacked in this order from the first substrate **21**. The insulating layers **221** to **229** have light transmittance and an insulating property. The material of each of the insulating layers **221** to **229** is an inorganic material containing silicon, such as silicon oxide and silicon oxynitride. Further, the insulating layer **221** is equivalent to a "second insulating layer", and the insulating layers **223** and **224** configure a first insulating layer **201**.

[0057] In the stacked body **22**, the transistors **23**, the scanning lines **241**, the signal lines **242**, the first light shielding portion **6**, and the third light shielding portion **7** are disposed. Further, in the stacked body **22**, relay electrodes **244**, **245**, **246**, **247**, **248**, and **249** are disposed. In addition, the second light shielding portion **210** is disposed on the first substrate **21**.

[0058] As described above, the first substrate **21** is constituted by, for example, a glass substrate or a quartz substrate. The first substrate **21** has a concave portion H**1**. The concave portion H**1** is a recess formed in the first substrate **21** and is formed for each transistor **23**. The concave portion H**1** is formed in the Y**1** direction which is a direction in which a semiconductor layer **231** to be described later extends.

[0059] The second light shielding portion **210** is disposed within the concave portion H**1**. The second light shielding portion **210** is formed, for example, using a damascene method. The second light shielding portion **210** is provided to prevent light from being incident on the semiconductor layer **231** of the transistor **23**. The first substrate **21** may not have the concave portion H**1**. In this case, the second light shielding portion **210** is disposed on a flat surface of the first substrate **21** which faces in the Z**1** direction. In addition, the second light shielding portion **210** overlaps the semiconductor layer **231** of the transistor **23** to be described below in plan view and is disposed between the first substrate **21** and the semiconductor layer **231**. The second light shielding portion **210** functions as a back surface light shielding portion disposed in a layer below the semiconductor layer **231**.

[0060] An intermediate layer **8** is disposed on the second light shielding portion **210**. The intermediate layer **8** is provided for each pixel P and includes a first intermediate portion **83** and a second intermediate portion **84**. The intermediate layer **8** is provided to adjust a distance between the second light shielding portion **210** and the first light shielding portion **6** to be described below.

[0061] The second transistor **23** is disposed on the insulating layer **221**. The transistor **23** includes the semiconductor layer **231**, a gate electrode **232**, and a gate insulating film **233**. The semiconductor layer **231** is disposed on the insulating layer **221** equivalent to a "second insulating layer". The insulating layer **221** is a layer between the semiconductor layer **231** and the second light shielding portion **210**. Furthermore, the gate electrode **232** is disposed on the insulating layer

222. The gate insulating film **233** is interposed between the gate electrode **232** and the semiconductor layer **231**. The region of the insulating layer **222** which corresponds to the gate electrode **232** in plan view is equivalent to the gate insulating film **233**.

[0062] FIG. **7** is a plan view of the transistor **23** in FIG. **5**. The transistor **23** shown in FIG. **7** has a lightly doped drain (LDD) structure. The semiconductor layer **231** extends in the Y**1** direction. The width direction of the semiconductor layer **231** is along the X-axis. The semiconductor layer **231** is disposed between the first substrate **21** and the gate electrode **232**. The semiconductor layer **231** has a drain region **231***a*, a source region **231***b*, a channel region **231***c*, a lightly doped drain region **231***d*, and a lightly doped source region **231***e*. The channel region **231***c* is located between the drain region **231***a* and the source region **231***b*. The lightly doped drain region **231***d* is located between the channel region **231***c* and the drain region **231***a*. The lightly doped source region **231***e* is located between the channel region **231***c* and the source region **231***b*. The semiconductor layer **231** is made of polysilicon, for example. A region other than the channel region **231***c* is doped with impurities that increase conductivity. The impurity concentration in the lightly doped drain region **231***d* is lower than the impurity concentration in the drain region **231***a*. The impurity concentration in the lightly doped source region **231***e* is lower than the impurity concentration in the source region **231***b*. For example, the transistor **23** does not need to have an LDD structure, and the lightly doped source region **231***e* and the lightly doped drain region **231***d* may be omitted.

[0063] The gate electrode **232** is made of polysilicon doped with impurities that increase conductivity, for example. Note that the gate electrode **232** may be made of a conductive material of a metal, metal oxide, and metal compound. The gate electrode **232** overlaps with the channel region **231***c* of the semiconductor layer **231** in plan view. In addition, the gate insulating film **233** is composed of a silicon oxide film deposited by a heat oxidation or chemical vapor deposition (CVD) method or the like.

[0064] As shown in FIG. **5**, a first light shielding portion **6** is disposed in the insulating layer **222**. The first light shielding portion **6** is formed, for example, using a damascene method. The first light shielding portion **6** is disposed in a through hole formed in the insulating layers **221** to **223**. The through hole includes a first concave portion R**1**, a second concave portion R**2**, and a third concave portion R**3**. In addition, the first light shielding portion **6** is directly coupled to the drain region **231***a* of the semiconductor layer **231**. Thus, the first light shielding portion **6** is a pixel potential.

[0065] As shown in FIG. **5**, the first light shielding portion **6** includes a first portion **61**, a second portion **62**, and a third portion **63**. The first portion **61** is disposed in the first concave portion R**1** described above. The second portion **62** is disposed in the second concave portion R**2**. The third portion **63** is disposed in the third concave portion R**3**. The first light shielding portion **6** functions as an upper light shielding portion disposed in a layer above the semiconductor layer **231**.

[0066] As described above, in the element substrate **2**, the second light shielding portion **210** is provided in the lower layer of the semiconductor layer **231**, and the first light shielding portion **6** is disposed in the upper layer of the semiconductor layer **231**. For this reason, a light shielding property can be improved compared to the related art.

[0067] As shown in FIG. **5**, the relay electrode **244** is disposed in the insulating layer **223**. The relay electrode **244** is electrically coupled to the source region **231***b* of the semiconductor layer **231** via a contact **271**. For example, the contact **271** is a contact plug that fills a hole that penetrates the insulating layers **222** and **223**. The relay electrode **244** and the contact **271** may be integrally formed of the same material, or may be formed of different materials.

[0068] As shown in FIG. **5**, the scanning line **241**, the relay electrode **245** and the relay electrode **246** are disposed on the insulating layer **224**. The scanning line **241** is electrically coupled to the gate electrode **232** via the third light shielding portion **7**. The relay electrode **245** is electrically coupled to the first light shielding portion **6** via a contact **272** that penetrates the insulating layer **224**. The relay electrode **246** is electrically coupled to the relay electrode **244** via a contact **273** that penetrates the insulating layer **224**. In addition, the first insulating layer **201** configured with the insulating layers **224** and **223** is located between the scanning line **241** and the gate electrode **232**.

[0069] As shown in FIG. **6**, the third light shielding portion **7** is disposed in the insulating layers **221** to **224**. The third light shielding portion **7** is formed using, for example, a damascene method. The third light shielding portion **7** is directly coupled to the scanning line **241** and the gate electrode **232**. Thus, the third light shielding portion **7** is at a gate potential.

[0070] The third light shielding portion **7** includes two fourth portions **71** and a fifth portion **72**. The two fourth portions **71** are spaced apart from each other, extend from the scanning line **241** to the second light shielding portion **210**, and are coupled thereto. Thus, the second light shielding portion **210** has the same potential as that of the gate electrode **232** and functions as a back gate. Each of the two fourth portions **71** is positioned on the outer side of the first portion **61** in a direction along the X-axis. The fifth portion **72** is disposed between the two fourth portions **71** and is coupled to the scanning line **241**.

[0071] The relay electrode **247** and the relay electrode **248** are disposed on the insulating layer **225**. The relay electrode **247** is electrically coupled to the relay electrode **246** via a contact **275** that penetrates the insulating layer **225**. The relay electrode **248** is electrically coupled to the relay electrode **245** via a contact **274** that penetrates the insulating layer **225**.

[0072] The signal line **242** is disposed on insulating layer **226**. The signal line **242** is electrically coupled to the relay electrode **247** via a contact **276** that penetrates the insulating layer **226**. Thus, the signal line **242** is electrically coupled to the source region **231***b* via the contact **276**, the relay electrode **247**, the contact **275**, the relay electrode **246**, the contact **273**, the relay electrode **244** and the contact **271**.

[0073] As shown in FIG. **6**, the relay electrode **249** is disposed on the insulating layer **226**. The relay electrode **249** is electrically coupled to the relay electrode **248** via a contact **277** that penetrates the insulating layer.

[0074] A capacitive element **24** is disposed on the insulating layer **227**. The capacitive element **24** includes a pair of electrodes **2401** and **2402** and a dielectric layer **2403**. The electrode **2401** is disposed on the insulating layer **227**. The electrode **2402** is disposed on the insulating layer **228**. The dielectric layer **2403** is disposed between the electrodes **2401** and **2402**. The dielectric layer **2403** contains, for example, HfO.sub.2, Al.sub.2O.sub.3, or the like. The electrode **2401** also serves as the constant potential line **243** in FIG. **2**. In addition, as shown in FIG. **6**, the electrode **2402** is electrically coupled to the relay electrode **249** via a contact **278** that penetrates the insulating layers **227** and **228**. Thus, as shown in FIGS. **5** or **6**, the electrode **2402** is electrically coupled to the drain region **231***a* via the contact **278**, the relay electrode **249**, the contact **277**, the relay electrode **248**, the contact **274**, the relay electrode **245**, the contact **272** and the first light shielding portion **6**.

[0075] As shown in FIG. **5**, the pixel electrode **25** is disposed on the insulating layer **229**. The pixel electrode **25** is electrically coupled to the electrode **2402** via a contact **279** that penetrates the insulating layer **229**.

[0076] Each of the above-mentioned scanning line **241**, signal line **242**, electrode **2401**, electrode **2402**, relay electrodes **244**, **245**, **246**, **247**, **248**, and **249** contains, for example, a metal such as tungsten (W), titanium (Ti), chromium (Cr), iron, and aluminum (Al), a metal nitride, a metal silicide, or the like. These may be single layers or multilayers. For example, these are constituted by a stacked body of an aluminum film and a titanium nitride film.

[0077] In addition, each of the above-mentioned contacts **271** to **279** contains, for example, a metal such as tungsten (W), titanium (Ti), chromium (Cr), iron (Fe), and aluminum (Al), a metal nitride, and a metal silicide. Each of the contacts **271** to **279** may be a single layer or a stacked layer. Each of the contacts **271** to **279** may be formed integrally with the electrode or wiring line to which it is coupled, or may be formed separately.

[0078] The configuration of the element substrate **2** shown in FIGS. **5** and **6** is merely one example. For example, the element substrate **2** may include capacitive elements other than the capacitive element **24**. In addition, the scanning line **241**, the signal line **242**, and the capacitive element **24** are arranged in this order in the Z**1** direction, but they do not have to be arranged side by side.

1D. First Light Shielding Portion **6**, Third Light Shielding Portion **7**, and Second Light Shielding Portion **210**

[0079] FIG. **8** is a plan view of the second light shielding portion **210** shown in FIG. **5**. The second light shielding portion **210** shown in FIG. **8** has an elongated shape of which the longitudinal direction is a direction along the Y-axis which is a direction in which the scanning line **241** extends in plan view.

[0080] The second light shielding portion **210** includes a wide portion **211**, a first narrow portion **212**, and a second narrow portion **213**. The wide portion **211** is positioned between the first narrow portion **212** and the second narrow portion **213**, and is wider than the first narrow portion **212** and the second narrow portion **213** in a width that is a length along the X-axis. The first narrow portion **212** extends from the wide portion **211** in the Y**1** direction. The second narrow portion **213** extends from the wide portion **211** in the Y**2** direction.

[0081] Referring to FIGS. **7** and **8**, the wide portion **211** overlaps the gate electrode **232** in plan view. The first narrow portion **212** overlaps the source region **231***b* in plan view. The second narrow portion **213** overlaps the drain region **231***a* in plan view.

[0082] FIGS. **9** and **10** are plan views of the first light shielding portion **6** and the third light shielding portion **7**, respectively. In FIG. **10**, in order to facilitate understanding, the first portion **61** is hatched, the second portion **62** is meshed, and the third portion **63** is dotted. The fourth portion **71** is hatched, and the fifth portion **72** is dotted.

[0083] As shown in FIG. **9**, the first light shielding portion **6** and the third light shielding portion **7** overlap the second light shielding portion **210** in plan view. The first light shielding portion **6** and the third light shielding portion **7** are disposed to be separated from each other. The third light shielding portion **7** is positioned outside the first light shielding portion **6** in plan view. The first light shielding portion **6** overlaps the drain region **231***a* and the lightly doped drain region **231***d* of the semiconductor layer **231** in plan view. Furthermore, the third light shielding portion **7** overlaps the gate electrode **232** and the channel region **231***c* of the semiconductor layer **231** in plan view.

[0084] As described above, the first light shielding portion **6** includes the first portion **61**, the second portion **62**, and the third portion **63**. Specifically, the first portion **61** includes a drain coupling portion **611** and two side surface light shielding portions **612**. The drain coupling portion **611** is a portion that overlaps the drain region **231***a* in plan view and is directly coupled to the drain region **231***a*. The drain coupling portion **611** is positioned in the Y**2** direction with respect to the second portion **62** and extends along the X-axis. The side surface light shielding portions **612** do not overlap the semiconductor layer **231** in plan view. The side surface light shielding portions **612** are positioned in the X**1** direction or the X**2** direction with respect to the second portion **62** and the third portion **63** in plan view, and extend substantially along the Y-axis.

[0085] The second portion **62** is positioned in the Y**2** direction with respect to the third portion **63** in plan view. The second portion **62** overlaps the lightly doped drain region **231***d* in plan view. The lightly doped drain region **231***d* is a region of the semiconductor layer **231** in which light leakage is most likely to occur. Thus, the second portion **62** overlaps the lightly doped drain region **231***d* in plan view, and thus it is possible to effectively curb the incidence of light on the lightly doped drain region **231***d* by the first light shielding portion **6**. Thus, it is possible to effectively curb light leakage.

[0086] As described above, in the present embodiment, the transistor **23** has an LDD structure. However, the lightly doped source region **231***e* and the lightly doped drain region **231***d* may be omitted. In this case, it is preferable that the second portion **62** overlap a bonding portion between the drain region **231***a* and the source region **231***b* in plan view. The bonding portion is a region where light leakage is likely to occur. Thus, the second portion **62** overlaps the bonding portion in plan view, and thus it is possible to effectively curb the incidence of light on the bonding portion by the first light shielding portion **6**. Thus, it is possible to effectively curb light leakage.

[0087] In addition, the third portion **63** overlaps the gate electrode **232** in plan view. For this reason, compared to a case where the third portion **63** does not overlap the gate electrode **232** in plan view, it is possible to curb the incidence of light on the semiconductor layer **231**, particularly the lightly doped drain region **231***d*.
[0088] As shown in FIG. **9**, each of the fourth portions **71** includes a portion extending along the Y-axis and a portion extending along the X-axis in plan view. The fifth portion **72** extends along the X-axis and is positioned between the two fourth portions **71** in plan view. As shown in FIGS. **9** and **10**, the fifth portion **72** overlaps the gate electrode **232** in plan view. Each of the two fourth portions **71** is positioned outside the first portion **61** in a direction along the X-axis in plan view, and does not overlap the gate electrode **232** in plan view.
[0089] The third light shielding portion **7** and the first portion **61** surround the lightly doped drain region **231***d* in plan view. For this reason, it is possible to effectively curb the incidence of light from a direction along the XY plane on the lightly doped drain region **231***d*.
[0090] Furthermore, the third light shielding portion **7** is positioned further outward from the first light shielding portion **6** in the X-axis direction. For this reason, as compared to a case where the third light shielding portion **7** is positioned on the inner side of the first light shielding portion **6** in the X-axis direction, it is possible to curb the influence of the third light shielding portion **7**, which is a gate potential, on the lightly doped drain region **231***d*. Thus, it is possible to curb an increase in an off-leak current. For this reason, the reduction in display quality due to the occurrence of black spots and the like can be curbed. Note that the off-leak current is a leakage current that flows when the transistor **23** is turned off.
[0091] Furthermore, as shown in FIG. **5**, the first portion **61** of the first light shielding portion **6** is directly coupled to the drain region **231***a*. For this reason, the potential of the first light shielding portion **6** and the potential of the drain region **231***a* of the semiconductor layer **231** are the same potential. Thus, even when the first light shielding portion **6** is brought close to the semiconductor layer **231**, problems are not likely to occur. Thus, it is easy to bring the first light shielding portion **6** close to the semiconductor layer **231**. For this reason, although each of the second portion **62** and the third portion **63** is separated from the semiconductor layer **231**, it is easy to bring them close to the semiconductor layer **231**. Since the first light shielding portion **6** can be brought close to the semiconductor layer **231**, a light shielding property of the first light shielding portion **6** can be improved compared to the related art.
[0092] As described above, the potential of the first light shielding portion **6** and the potential of the second light shielding portion **210** are different from each other. Since these potentials are different from each other, it is possible to respectively set the potential of the first light shielding portion **6** and the potential of the second light shielding portion **210**. For this reason, it is possible to reduce the electrical adverse effect of each of the first light shielding portion **6** and the second light shielding portion **210** on the semiconductor layer **231**.
[0093] Specifically, the first light shielding portion **6** is directly coupled to the drain region **231***a* and is electrically coupled to the drain region **231***a*. In addition, the second light shielding portion **210** is electrically coupled to the gate electrode **232** and overlaps the semiconductor layer **231** in plan view. Since the first light shielding portion **6** is electrically coupled to the drain region **231***a*, even when the first light shielding portion **6** is brought close to the drain region **231***a*, an electrical failure is unlikely to occur. Thus, the first light shielding portion **6** can be brought close to the lightly doped drain region **231***d*. Accordingly, the first light shielding portion **6** can be brought close to the semiconductor layer **231**, and thus a light shielding property of the first light shielding portion **6** can be improved compared to the related art. On the other hand, the second light shielding portion **210** is electrically coupled to the gate electrode **232**, and thus the second light shielding portion **210** can be used as a back gate.
[0094] The first light shielding portion **6** is at a pixel potential and not at a gate potential. Thus, even when the first light shielding portion **6** is disposed near the semiconductor layer **231**, it is

possible to curb the influence of the gate potential on the semiconductor layer **231**. More specifically, the increase in off-leak current due to the gate potential coming closer to the region other than the channel region **231**c of the semiconductor layer **231** can be curbed. In this manner, the reduction in display quality due to the occurrence of black spots and the like can be curbed.

[0095] Examples of materials for the first light shielding portion **6**, the second light shielding portion **210**, and the third light shielding portion **7** include metals such as tungsten (W), titanium (Ti), chromium (Cr), iron (Fe), and aluminum (Al), metal nitrides, and metal silicides. Among these, it is preferable that the first light shielding portion **6** and the third light shielding portion **7** contain tungsten. Among various metals, tungsten is excellent in heat resistance, and its optical density (OD) value is unlikely decrease through heat treatment during manufacturing, for example. Thus, the first light shielding portion **6**, the second light shielding portion **210**, and the third light shielding portion **7** contain tungsten, and thus it is possible to particularly effectively prevent light from being incident on the semiconductor layer **231** by the first light shielding portion **6**, the second light shielding portion **210**, and the third light shielding portion **7**. Furthermore, the materials of the first light shielding portion **6**, the second light shielding portion **210**, and the third light shielding portion **7** may be the same as or different from each other.

[0096] FIG. **11** is a cross-sectional view taken along line A**3**-A**3** in FIG. **9**. FIG. **12** is a cross-sectional view taken along line A**4**-A**4** in FIG. **9**.

[0097] As shown in FIGS. **11** and **12**, the two side surface light shielding portions **612** of the first portion **61** are disposed on the outer side of the semiconductor layer **231** in the X-axis direction. That is, the two side surface light shielding portions **612** are disposed on both sides of the semiconductor layer **231** in the width direction. For this reason, it is possible to curb the incidence of light on the semiconductor layer **231** in the X**1** direction and the X**2** direction by the two side surface light shielding portions **612**.

[0098] Further, as shown in FIG. **5**, FIG. **11**, or FIG. **12**, the side surface light shielding portion **612** of the first portion **61** extends from the first insulating layer **201**, which is a layer between the scanning line **241** and the gate electrode **232**, to the insulating layer **221**, which is a “second insulating layer”. As shown in FIG. **5**, the drain coupling portion **611** of the first portion **61** extends from the first insulating layer **201** to the drain region **231**a. In addition, as shown in FIG. **12**, the second portion **62** is provided between the semiconductor layer **231** and the insulating layer **225**, which is a layer in which the scanning line **241** is provided. As shown in FIG. **11**, the third portion **63** is provided between the gate electrode **232** and the insulating layer **225**.

[0099] A length L**3** of the third portion **63** in the Z**2** direction, a length L**2** of the second portion **62** in the Z**2** direction, and a length L**1** of the first portion **61** in the Z**2** direction are in ascending order. The length L**1** is the longest. Thus, the lengths L**1**, L**2**, and L**3** satisfy a relationship of L**3**<L**2**<L**1**. Each of the lengths L**1**, L**2**, and L**3** is a maximum length. Thus, the length L**1** is the length of the side surface light shielding portion **612**.

[0100] According to the configuration of the first light shielding portion **6** in which the lengths L**1**, L**2**, and L**3** satisfy a relationship of L**3**<L**2**<L**1**, the first light shielding portion **6** can be disposed near the semiconductor layer **231**. For this reason, it is possible to more effectively curb the incidence of light on the semiconductor layer **231** than the related art. Thus, it is possible to prevent the operation of the transistor **23** from becoming unstable. As a result, it is possible to curb a concern that display defects such as uneven brightness occur. Thus, it is possible to curb deterioration in display quality.

[0101] In addition, other wiring lines and electrodes are not interposed between the first light shielding portion **6** and the semiconductor layer **231**. For this reason, it is possible to bring the first light shielding portion **6** close to the semiconductor layer **231**. Thus, as compared with a case where wiring lines and electrodes are interposed between the first light shielding portion **6** and the semiconductor layer **231**, it is possible to curb the incidence of light on the semiconductor layer **231** in the Z**1** direction by the first light shielding portion **6**.

[0102] In addition, as shown in FIG. **5**, the upper surfaces of the first portion **61**, the second portion **62**, and the third portion **63** are flush with each other. Thus, a distance D**1** between the first portion **61** and the semiconductor layer **231**, a distance D**2** between the second portion **62** and the semiconductor layer **231**, and a distance D**3** between the third portion **63** and the semiconductor layer **231** are in ascending order. The length D**3** is the longest. Thus, the lengths D**1**, D**2**, and D**3** satisfy a relationship of D**1**<D**2**<D**3**. The length D**3** is 0 (zero). For this reason, the second portion **62** is closer to the semiconductor layer **231** than the third portion **63** in a direction along the Z-axis. The first light shielding portion **6** includes the second portion **62**, it is possible to bring the first light shielding portion **6** close to the semiconductor layer **231**. Thus, the incidence of light on the semiconductor layer **231** can be more effectively exhibited.
[0103] In particular, the surface of the second portion **62** which faces the semiconductor layer **231** is preferably provided at a position between the surface of the gate electrode **232** which faces the semiconductor layer **231** and the surface of the gate electrode **232** which faces the scanning line **241** in a direction along the Z-axis. The surface of the second portion **62** which faces the semiconductor layer **231** is provided at such a position, and thus a distance between the second portion **62** and the semiconductor layer **231** can be significantly reduced. Furthermore, the second portion **62** overlaps the lightly doped drain region **231***d* in plan view, and thus the second portion **62** can be brought significantly close to the lightly doped drain region **231***d*. That is, a distance between the second portion **62** and the lightly doped drain region **231***d* can be made significantly smaller than that in the related art. For this reason, it is possible to prevent light from being incident on the semiconductor layer **231**, particularly on the lightly doped drain region **231***d* in the Z**1** direction. Curbing the incidence of light in the Z**1** direction is particularly effective in a mode in which light is incident from the opposing substrate **3** as in the present embodiment.
[0104] Examples of materials for the first light shielding portion **6**, the second light shielding portion **210**, and the third light shielding portion **7** include metals such as tungsten (W), titanium (Ti), chromium (Cr), iron (Fe), and aluminum (Al), metal nitrides, and metal silicides. Among these, it is preferable that the first light shielding portion **6** and the third light shielding portion **7** contain tungsten. Among various metals, tungsten is excellent in heat resistance, and its optical density (OD) value is unlikely decrease through heat treatment during manufacturing, for example. Thus, the first light shielding portion **6**, the second light shielding portion **210**, and the third light shielding portion **7** contain tungsten, and thus it is possible to particularly effectively prevent light from being incident on the semiconductor layer **231** by the first light shielding portion **6**, the second light shielding portion **210**, and the third light shielding portion **7**. Furthermore, the materials of the first light shielding portion **6**, the second light shielding portion **210**, and the third light shielding portion **7** may be the same as or different from each other.
[0105] For example, the third concave portion R**3** may be omitted. In this case, the third portion **63** may be omitted.
1E. Intermediate Layer **8**
[0106] As shown in FIGS. **11** and **12**, the intermediate layer **8** is disposed between the first portion **61** of the first light shielding portion **6** and the second light shielding portion **210** so as to be in contact therewith. The intermediate layer **8** has an insulating property. The intermediate layer **8** contains a material different from the materials of the first insulating layer **201** and the insulating layer **221** serving as a “second insulating layer”. That is, the intermediate layer **8** is different from the materials of the insulating layers **221** to **223**. By providing such an intermediate layer **8**, it is possible to avoid electrical coupling between the first light shielding portion **6** and the second light shielding portion **210**. Furthermore, by using the intermediate layer **8** as an etching stopper at the time of manufacturing the element substrate **2**, it is possible to reduce the occurrence of a variation in a distance between the first portion **61** of the first light shielding portion **6** and the second light shielding portion **210** for each pixel P. In addition, it is possible to reduce variations in light shielding properties for each electro-optical device **100**. The plane area of the intermediate layer **8**

is larger than the plane areas of the first portion **61** and the second light shielding portion **210**.
[0107] Since such an intermediate layer **8** is provided, it is possible to curb occurrence of a variation in light resistance of the semiconductor layer **231** due to a processing variation in the side surface light shielding portion **612** in the plurality of electro-optical devices **100**. Specifically, when the end portion of the side surface light shielding portion **612** in the Z**2** direction is positioned in a layer above the second light shielding portion **210**, a distance between the side surface light shielding portion **612** and the second light shielding portion **210** is likely to vary for each pixel P due to manufacturing errors or the like. In the present embodiment, the intermediate layer **8** is provided between each side surface light shielding portion **612** and the second light shielding portion **210**. For this reason, by using the intermediate layer **8** as an etching stopper, it is possible to reduce the occurrence of a variation in a distance between the first portion **61** of the first light shielding portion **6** and the second light shielding portion **210** for each pixel P. Thus, it is possible to effectively curb a decrease in yield. Since it is possible to maintain a constant light shielding shape as compared with the related art, it is possible to stabilize the quality of the electro-optical device **100**.
[0108] It is preferable that an etching rate of the intermediate layer **8** with respect to a fluorine-based etchant be lower than those of the insulating layer **221** and the first insulating layer **201**. Thereby, the intermediate layer **8** can be suitably used as an etching stopper.
[0109] The insulating layer **221** and the first insulating layer **201** contain silicon oxide (SiOx) or silicon oxynitride (SiON). The intermediate layer **8** contains silicon nitride (SiN). In this case, the etching rate of the intermediate layer **8** with respect to a fluorine-based etchant is lower than those of the insulating layer **221** and the first insulating layer **201**. For this reason, the intermediate layer **8** can be suitably used as an etching stopper.
[0110] In addition, a thickness D**8** of the intermediate layer **8** is smaller than a thickness L**1** of the first portion **61** of the first light shielding portion **6**. Further, the thickness D**8** of the intermediate layer **8** is smaller than a thickness L**4** of the second light shielding portion **210**. Since the thickness D**8** is smaller than the thickness L**4**, a distance between the first light shielding portion **6** and the second light shielding portion **210** can be reduced as compared with when the thickness D**8** is larger than the thickness L**4**. For this reason, it is possible to curb penetration of light into the semiconductor layer **231** from between the first light shielding portion **6** and the second light shielding portion **210**. Thus, it is possible to reduce the occurrence of a variation in each pixel P due to the provision of the intermediate layer **8** and to particularly effectively exhibit the effect of curbing the penetration of light into the semiconductor layer **231**.
[0111] The thickness D**8** of the intermediate layer **8** is not particularly limited and is, for example, 1000 Å or more and 2000 Å or less. When the thickness is within the above range, it is possible to curb deterioration in a light shielding property due to an increase in the distance between the first light shielding portion **6** and the second light shielding portion **210** while allowing the intermediate layer **8** to suitably function as an etching stopper, as compared with a case where the thickness falls outside the range.
[0112] FIG. **13** is a plan view of the intermediate layer **8** shown in FIG. **12**. As shown in FIG. **13**, the intermediate layer **8** includes a first intermediate portion **83** and a second intermediate portion **84**. The second intermediate portion **84** is disposed in the Y**1** direction with respect to the first intermediate portion **83** in plan view. The intermediate layer **8** does not overlap the third light shielding portion **7** in plan view and is not coupled to the third light shielding portion **7**. Specifically, the third light shielding portion **7** is disposed between the first intermediate portion **83** and the second intermediate portion **84** in plan view. In addition, the first intermediate portion **83** overlaps the first light shielding portion **6** in plan view.
[0113] The intermediate layer **8** is also provided in the light shielding region A**12** described above, but is not provided in the opening region A**11**. Thus, even when the intermediate layer **8** is formed of a material that is not excellent in light transmittance, such as silicon nitride, the presence of the

intermediate layer **8** makes it possible to avoid a decrease in light transmittance in the opening region A**11**. When the intermediate layer **8** is light transmissive, a portion of the intermediate layer **8** may be provided in the opening region A**11**. In addition, the intermediate layer **8** may be disposed at least between the first portion **61** and the second light shielding portion **210**. Thus, the second intermediate portion **84** may be omitted.

1F. Method of Manufacturing Intermediate Layer **8** and First Light Shielding Portion **6**

[0114] The intermediate layer **8** and the first light shielding portion **6** described above are manufactured, for example, by the following method. The manufacturing method for these is not limited to the following method.

[0115] FIGS. **14** and **15** are diagrams showing a method of manufacturing the intermediate layer **8** shown in FIG. **12**. As shown in FIG. **14**, an insulating film **8***a* is formed at the first substrate **21** to cover the second light shielding portion **210**. The insulating film **8***a* is made of a material having an insulating property such as silicon nitride. The insulating film **8***a* is formed by, for example, a vapor deposition method such as chemical vapor deposition (CVD) or a sputtering method. Next, the intermediate layer **8** shown in FIG. **15** is formed by removing a portion of the insulating film **8***a* by etching.

[0116] FIG. **16** is a cross-sectional view showing a method of manufacturing the first concave portion R**1**. FIG. **17** is a plan view showing a method of manufacturing the first concave portion R**1**. As shown in FIGS. **16** and **17**, after the insulating layer **223** is formed, the first concave portion R**1** penetrating the insulating layers **221** to **223** is formed by etching. For example, when the insulating layers **221** to **223** are made of an inorganic material containing silicon, portions of the insulating layers **221** to **223** are removed by etching using a fluorine-based etchant. At this time, when the etching rate of the intermediate layer **8** with respect to the etchant is slower than those of the insulating layers **221** to **223**, the intermediate layer **8** functions as an etching stopper. Thus, it is possible to reduce the occurrence of a variation in the depth of the first recess portion R**1** for each pixel P.

[0117] FIG. **18** is a cross-sectional view showing a method of manufacturing the second concave portion R**2**. FIG. **19** is a plan view showing a method of manufacturing the second concave portion R**2**. Next, as shown in FIGS. **18** and **19**, the second concave portion R**2** is formed in the insulating layer **223** by etching. As described above, for example, when the insulating layer **223** is made of an inorganic material containing silicon, a portion of the insulating layer **223** is removed by etching using a fluorine-based etchant.

[0118] FIG. **20** is a cross-sectional view showing a method of manufacturing the third concave portion R**3**. Next, as shown in FIG. **20**, the third concave portion R**3** is formed in the insulating layer **223** by etching. As described above, for example, when the insulating layer **223** is made of an inorganic material containing silicon, a portion of the insulating layer **223** is removed by etching using a fluorine-based etchant.

[0119] The order of manufacturing the first concave portion R**1**, the second concave portion R**2**, and the third concave portion R**3** is not limited to the above-described order. For example, portions of the first concave portion R**1**, the second concave portion R**2**, and the third concave portion R**3** may be formed at the same time.

[0120] FIG. **21** is a diagram showing a method of manufacturing the first light shielding portion **6**. As shown in FIG. **21**, the first light shielding portion **6** is formed by filling the first concave portion R**1**, the second concave portion R**2**, and the third concave portion R**3** with a conductive material such as tungsten.

2. Second Embodiment

[0121] In a second embodiment described below, for elements having the same operations or functions as those in the first embodiment, the reference numerals used in the description of the first embodiment will be used, and detailed descriptions thereof will be omitted as appropriate.

[0122] FIGS. **22** and **23** are cross-sectional views showing an intermediate layer **8**A in the second embodiment. FIG. **24** is a plan view showing a second layer **82** of the intermediate layer **8**A shown in FIG. **22**.

[0123] As shown in FIGS. **22** and **23**, the intermediate layer **8**A of the present embodiment includes a first layer **81** and the second layer **82**. The first layer **81** and the second layer **82** are stacked in order from a second light shielding portion **210**. The second layer **82** is in contact with the second light shielding portion **210**. The first layer **81** is a layer having an insulating property. Furthermore, in the present embodiment, the first layer **81** has light transmittance. The first layer **81** is uniformly provided in both a light shielding region A**12** and an opening region A**11**. The second layer **82** contains a material different from those of insulating layers **221** to **223**. In the present embodiment, the second layer **82** contains a metal. The etching rate of the second layer **82** with respect to a fluorine-based etchant is lower than the etching rates of the insulating layers **221** to **223**. The second layer **82** is in contact with a side surface light shielding portion **612** of a first light shielding portion **6**. In addition, the second layer **82** is provided in the light shielding region A**12**. For this reason, the presence of the second layer **82** makes it possible to avoid a decrease in light transmittance in the opening region A**11**.

[0124] By providing the intermediate layer **8**A including the first layer **81** having an insulating property, it is possible to avoid electrical coupling between the first light shielding portion **6** and the second light shielding portion **210**. Further, since the intermediate layer **8**A includes the second layer **82** containing a material different from those of the insulating layers **221** to **223**, it is possible to reduce the occurrence of a variation in a distance between the first portion **61** of the first light shielding portion **6** and the second light shielding portion **210** for each pixel P by using the intermediate layer **8**A as an etching stopper at the time of manufacturing an element substrate **2**. Thus, it is possible to reduce a variation in light shielding performance in an electro-optical device **100**.

[0125] The order of stacking the first layer **81** and the second layer **82** may be opposite to that in the example of FIG. **22**. That is, the second layer **82** and the first layer **81** may be stacked in order from the second light shielding portion **210**. For example, the first layer **81** is made of an

[0126] insulating material containing silicon such as silicon oxide or silicon oxynitride. For example, the first layer **81** is made of the same material as those of the insulating layers **221** to **223**. Since the first layer **81** is made of the same material as those of the insulating layers **221** to **223**, the first layer **81** can be easily formed as compared with a case where the first layer **81** is made of a different material, and even when the first layer **81** is provided in the opening region A**11**, it is possible to curb a decrease in light transmittance in the opening region A**11**.

[0127] On the other hand, for example, the second layer **82** is formed of a metal such as tungsten, titanium, chromium, iron, or aluminum, a metal nitride, or a metal silicide. For example, the second layer **82** is formed of the same material as that of the first light shielding portion **6**. Further, the second layer **82** may be formed of silicon nitride (SiN).

[0128] A thickness D**8** of the intermediate layer **8**A is the same as that in the first embodiment described above. A thickness of the second layer **82** of the intermediate layer **8**A is not particularly limited and is 1000 Å or more and 2000 Å or less. The thickness of the second layer **82** is within the above range, and thus it is possible to more easily perform film formation than when the thickness of the second layer **82** falls outside the range, and to curb a decrease in a light shielding property due to an increase in a distance between the first light shielding portion **6** and the second light shielding portion **210**.

[0129] The thickness of the first layer **81** is not particularly limited, but is 500 Å or more and 1000 Å or less. When the thickness of the first layer **81** is within the above-described range, it is possible to curb a decrease in a light shielding property due to an increase in a distance between the first light shielding portion **6** and the second light shielding portion **210** while curbing a decrease in the light transmittance of the opening region A**11** as compared with a case where the thickness falls outside the range.

[0130] As shown in FIGS. **23** and **24**, the second layer **82** includes a first intermediate portion **821** and a second intermediate portion **822**. The second intermediate portion **822** is disposed in the Y**1** direction with respect to the first intermediate portion **821**. The second layer **82** does not overlap a third light shielding portion **7** in plan view and is not coupled to the third light shielding portion **7**. Specifically, the third light shielding portion **7** is disposed between the first intermediate portion **821** and the second intermediate portion **822** in plan view. In addition, the first intermediate portion **821** overlaps the first light shielding portion **6** in plan view. The second intermediate portion **822** may be omitted.

3. Modification Example

[0131] The embodiments illustrated above may be modified in various ways. Specific modifications that can be applied to the above-described embodiments are illustrated below. Two or more aspects arbitrarily selected from examples below may be combined appropriately as long as contradiction is not caused.

[0132] Each of the intermediate layers **8** and **8**A may further include another layer.

[0133] In the above description, the third light shielding portion **7** is coupled to the second light shielding portion **210**, but may not be coupled. In addition, the third light shielding portion **7** may be omitted.

[0134] In each of the embodiments described above, the electro-optical device **100** using an active matrix scheme is illustrated, but the present embodiment is not limited thereto and a drive scheme for the electro-optical device **100** may be, for example, a passive matrix scheme.

[0135] The driving scheme of the "electro-optical device" is not limited to a vertical electric field scheme, but may be a transverse electric field scheme. An example of the transverse electric field scheme may include an In Plane Switching (IPS) mode. Further, examples of the vertical electric field scheme may include a twisted nematic (TN) mode, a vertical alignment (VA) mode, a PVA mode, and an optically compensated bend (OCB) mode.

4. Electronic Apparatus

[0136] The electro-optical device **100** can be used in various electronic apparatuses.

[0137] FIG. **25** is a perspective view showing a personal computer **2000** that is an example of the electronic apparatus. The personal computer **2000** includes the electro-optical device **100** that displays various images, a body unit **2010** in which a power switch **2001** and a keyboard **2002** are installed, and a control unit **2003**. The control unit **2003** includes, for example, a processor and a memory, and controls an operation of the electro-optical device **100**.

[0138] FIG. **26** is a plan view showing a smartphone **3000** that is an example of the electronic apparatus. The smartphone **3000** includes an operation button **3001**, an electro-optical device **100** that displays various images, and a control unit **3002**. Screen content displayed on the electro-optical device **100** is changed according to an operation of the operation button **3001**. The control unit **3002** includes, for example, a processor and a memory, and controls an operation of the electro-optical device **100**.

[0139] FIG. **27** is a schematic diagram showing a projector that is an example of the electronic apparatus. A projection type display apparatus **4000** is, for example, a three-panel projector. An electro-optical device **1***r* is an electro-optical device **100** corresponding to a red display color, an electro-optical device **1***g* is an electro-optical device **100** corresponding to a green display color, and an electro-optical device **1***b* is an electro-optical device **100** corresponding to a blue display color. That is, the projection type display apparatus **4000** includes three electro-optical devices **1***r*, **1***g*, and **1***b* corresponding to respective red, green, and blue display colors. A control unit **4005** includes, for example, a processor and a memory, and controls an operation of the electro-optical device **100**.

[0140] An illumination optical system **4001** supplies a red component r of the light emitted from an illumination apparatus **4002**, which is a light source, to the electro-optical device **1***r*, a green component g to the electro-optical device **1***g*, and a blue component b to the electro-optical device

1*b*. Each of the electro-optical devices **1***r*, **1***g*, and **1***b* functions as an optical modulator such as a light valve that modulates each monochromatic light beam supplied from the illumination optical system **4001** according to a displayed image. A projection optical system **4003** combines the light emitted from the respective electro-optical devices **1***r*, **1***g*, and **1***b* and projects the combined light onto a projection surface **4004**.
[0141] The electronic apparatus includes the electro-optical device **100** described above and a control unit **2003**, **3002**, or **4005**. The electro-optical device **100** described above has an excellent light shielding property because of the light shielding portion **6** of the semiconductor layer **231**, and thus the destabilization of the operation of the transistor **23** is curbed. In this manner, the concern of the occurrence of display failures is curbed. Thus, with the electro-optical device **100**, the display quality of the personal computer **2000**, the smartphone **3000**, or the projection type display apparatus **4000** can be increased.
[0142] An electronic apparatus to which the electro-optical device of the present disclosure is applied is not limited to the illustrated device, and examples thereof may include a personal digital assistant (PDA), a digital still camera, a television, a video camera, a car navigation apparatus, an in-vehicle display, an electronic notebook, an electronic paper, a calculator, a word processor, a workstation, a videophone, and a point of sale (POS) terminal. Further, examples of the electronic apparatus to which the present disclosure is applied may include a printer, a scanner, a copier, a video player, and a device including a touch panel.
[0143] Although the present disclosure has been described above based on the preferred embodiments, the present disclosure is not limited to the above-described embodiments. Further, a configuration of the respective portions of the present disclosure can be replaced with any configuration that performs the same function as that in the embodiment described above, or any configuration can be added.
[0144] Further, in the above description, the liquid crystal display device has been described as an example of the electro-optical device of the present disclosure, but the electro-optical device of the present disclosure is not limited thereto. For example, the electro-optical device of the present disclosure can be applied to an image sensor, or the like.

Claims

1. An electro-optical device comprising: a substrate; a transistor including a gate electrode disposed in a first direction with respect to the substrate and a semiconductor layer disposed between the substrate and the gate electrode; a scanning line disposed in the first direction with respect to the gate electrode and electrically coupled to the gate electrode; a first light shielding portion disposed in the first direction with respect to the semiconductor layer and overlapping the semiconductor layer in plan view when viewed in the first direction; and a second light shielding portion disposed in a second direction opposite to the first direction with respect to the semiconductor layer and overlapping the semiconductor layer in plan view, wherein a potential of the first light shielding portion and a potential of the second light shielding portion are different from each other, the first light shielding portion includes a first portion that is provided on both sides of the semiconductor layer in a width direction and extends from a first insulating layer between the scanning line and the gate electrode to a second insulating layer between the semiconductor layer and the second light shielding portion, an intermediate layer is disposed between the first portion and the second light shielding portion to be in contact with the first portion and the second light shielding portion, and the intermediate layer includes a layer having an insulating property and contains a material different from materials of the first insulating layer and the second insulating layer.
2. The electro-optical device according to claim 1, wherein the first light shielding portion further includes a second portion and a third portion, the second portion is provided between the semiconductor layer and a layer in which the scanning line is provided, the third portion is

provided between the gate electrode and a layer in which the scanning line is provided, and a length of the third portion in the second direction, a length of the second portion in the second direction, and a length of the first portion in the second direction are in ascending order, and the length of the first portion in the second direction is longest.
3. The electro-optical device according to claim 2, wherein a distance between the first portion and the semiconductor layer, a distance between the second portion and the semiconductor layer, and a distance between the third portion and the semiconductor layer are in ascending order, and the distance between the third portion and the semiconductor layer is the longest.
4. The electro-optical device according to claim 1, wherein the semiconductor layer includes a drain region, a source region, and a channel region provided between the drain region and the source region, the first light shielding portion is electrically coupled to the drain region and includes a portion overlapping the drain region in plan view, and the second light shielding portion is electrically coupled to the gate electrode.
5. The electro-optical device according to claim 1, wherein the intermediate layer has a lower etching rate with respect to a fluorine-based etchant than those of the first insulating layer and the second insulating layer.
6. The electro-optical device according to claim 1, wherein the intermediate layer contains silicon nitride.
7. The electro-optical device according to claim 1, wherein the intermediate layer includes a first layer containing a metal and a second layer having an insulating property.
8. The electro-optical device according to claim 7, wherein the first layer includes the same material as that of the second insulating layer.
9. The electro-optical device according to claim 1, wherein the intermediate layer has a thickness larger than a thickness of the second light shielding portion.
10. An electronic apparatus comprising: the electro-optical device according to claim 1; and a control unit configured to control an operation of the electro-optical device.